The F12 special term and the substituted final formula

generated by `f12_special_term.py`

The stage-4 space restriction killed every term carrying a t_2 amplitude whose two virtual indices both sit on the operator $\hat{E}$ (the free, external indices of the equation). In the CF-F12 construction those amplitudes are not independent: they are the fixed F12 geminal amplitudes of the SP ansatz,

$$\tilde{t}_{ij}^{ab} = \tfrac{3}{8}\langle ab|\hat{Q}_{12} f_{12}|ij\rangle + \tfrac{1}{8}\langle ab|\hat{Q}_{12} f_{12}|ji\rangle, \tag{1}$$

with a, b over the complete virtual space and $\hat{Q}_{12}$ the strong-orthogonality projector (the geminal vanishes with both indices OBS-vir, which is what made the stage-4 rule exact). The killed terms are the contribution of these fixed geminal amplitudes to the transcorrelated Hamiltonian – the *F12 special term*. This stage brings them back: the killed three-body terms are folded back to two-/one-body pieces with the grouped cumulant template of stage 6 (a density matrix with a virtual index is zero; the pure-density scalar part always vanishes), and the result is substituted into the cumulant-expanded blocks of stage 6.

1 The F12 special terms of each block

Only the blocks with killed terms are shown. The killed terms come first; when a block has killed three-body terms, their cumulant-folded form follows, then the special block (killed terms and folded pieces merged).

$\left[\hat{h},\, \hat{T}_2\right]$ **(1 killed terms)**

killed by the space restriction:

$$\left[\hat{h},\, \hat{T}_2\right] = -h_{p_1}^{i_2}\, t_{i_1 i_2}^{a_1 a_2}\, \hat{E}_{i_1 p_1}^{a_1 a_2} \tag{2}$$

$\left[\hat{V},\, \hat{T}_2\right]$ **(2 killed terms)**

killed by the space restriction:

$$\left[\hat{V},\, \hat{T}_2\right] = -\tfrac{1}{2}\, g_{p_1 p_2}^{i_1 i_2}\, t_{i_1 i_2}^{a_1 a_2}\, \hat{E}_{p_1 p_2}^{a_1 a_2} \tag{3}$$

$$- g_{p_2 p_3}^{i_2 p_1}\, t_{i_1 i_2}^{a_1 a_2}\, \hat{E}_{i_1 p_2 p_3}^{a_1 a_2 p_1} \tag{4}$$

the 1 three-body terms fold into 3 terms by the cumulant template:

$$\text{3-body part} = -D_{p_3}^{p_2}\, g_{p_1 p_3}^{i_2 p_2}\, t_{i_1 i_2}^{a_1 a_2}\, \hat{E}_{i_1 p_1}^{a_1 a_2} \tag{5}$$

$$+ \tfrac{1}{2}\, D_{p_3}^{p_2}\, g_{p_3 p_1}^{i_2 p_2}\, t_{i_1 i_2}^{a_1 a_2}\, \hat{E}_{i_1 p_1}^{a_1 a_2} \tag{6}$$

$$+ \tfrac{1}{2}\, D^{i_1}_{p_3} g^{i_2 p_3}_{p_1 p_2} t^{a_1 a_2}_{i_2 i_1} \hat{E}^{a_1 a_2}_{p_1 p_2} \tag{7}$$

the special block (killed terms and folded pieces merged, 4 terms):

$$\left[\hat{V},\, \hat{T}_2\right]^{F12} = -D^{p_2}_{p_3} g^{i_2 p_2}_{p_1 p_3} t^{a_1 a_2}_{i_1 i_2} \hat{E}^{a_1 a_2}_{i_1 p_1} \tag{8}$$

$$+ \tfrac{1}{2}\, D^{p_2}_{p_3} g^{i_2 p_2}_{p_3 p_1} t^{a_1 a_2}_{i_1 i_2} \hat{E}^{a_1 a_2}_{i_1 p_1} \tag{9}$$

$$+ \tfrac{1}{2}\, D^{i_1}_{p_3} g^{i_2 p_3}_{p_1 p_2} t^{a_1 a_2}_{i_2 i_1} \hat{E}^{a_1 a_2}_{p_1 p_2} \tag{10}$$

$$- \tfrac{1}{2}\, g^{i_1 i_2}_{p_1 p_2} t^{a_1 a_2}_{i_1 i_2} \hat{E}^{a_1 a_2}_{p_1 p_2} \tag{11}$$

$\left[\left[\hat{F},\, \hat{T}_1\right],\, \hat{T}_2\right]$ **(1 killed terms)**

killed by the space restriction:

$$\left[\left[\hat{F},\, \hat{T}_1\right],\, \hat{T}_2\right] = -f^{a_3}_{i_3} t^{a_1 a_2}_{i_1 i_3} t^{a_3}_{i_2} \hat{E}^{a_1 a_2}_{i_1 i_2} \tag{12}$$

$\left[\left[\hat{F},\, \hat{T}_2\right],\, \hat{T}_1\right]$ **(1 killed terms)**

killed by the space restriction:

$$\left[\left[\hat{F},\, \hat{T}_2\right],\, \hat{T}_1\right] = -f^{a_3}_{i_3} t^{a_1 a_2}_{i_1 i_3} t^{a_3}_{i_2} \hat{E}^{a_1 a_2}_{i_1 i_2} \tag{13}$$

$\left[\left[\hat{F},\, \hat{T}_2\right],\, \hat{T}_2\right]$ **(1 killed terms)**

killed by the space restriction:

$$\left[\left[\hat{F},\, \hat{T}_2\right],\, \hat{T}_2\right] = -2\, f^{a_4}_{i_4} t^{a_1 a_2}_{i_1 i_4} t^{a_4 a_3}_{i_2 i_3} \hat{E}^{a_1 a_2 a_3}_{i_1 i_2 i_3} \tag{14}$$

the 1 three-body terms fold into 0 terms by the cumulant template:

$$\text{3-body part} = 0 \quad \text{(no template group carries all three virtual indices)} \tag{15}$$

the special block (killed terms and folded pieces merged, 0 terms):

$$\left[\left[\hat{F},\, \hat{T}_2\right],\, \hat{T}_2\right]^{F12} = 0 \tag{16}$$

$\left[\left[\hat{F},\, \hat{T}_1\right]^{\dagger},\, \hat{T}_2\right]$ **(2 killed terms)**

killed by the space restriction:

$$\left[\left[\hat{F},\, \hat{T}_1\right]^{\dagger},\, \hat{T}_2\right] = f^{i_2}_{i_3} t^{a_1 a_2}_{i_2 i_1} t^{a_3}_{i_3} \hat{E}^{a_1 a_2}_{a_3 i_1} \tag{17}$$

$$- f^{a_3}_{p_1} t^{a_1 a_2}_{i_1 i_2} t^{a_3}_{i_2} \hat{E}^{a_1 a_2}_{i_1 p_1} \tag{18}$$

$\left[\left[\hat{F},\,\hat{T}_2\right]^{\dagger},\,\hat{T}_1\right]$ **(2 killed terms)**

killed by the space restriction:

$$\left[\left[\hat{F},\,\hat{T}_2\right]^{\dagger},\,\hat{T}_1\right] = f^{i_2}_{i_3} t^{a_1}_{i_2} t^{a_2 a_3}_{i_3 i_1} \hat{E}^{a_1 i_1}_{a_2 a_3} \tag{19}$$

$$+ f^{i_1}_{p_1} t^{a_1}_{i_2} t^{a_2 a_3}_{i_2 i_1} \hat{E}^{a_1 p_1}_{a_2 a_3} \tag{20}$$

$\left[\left[\hat{F},\,\hat{T}_2\right]^{\dagger},\,\hat{T}_2\right]$ **(6 killed terms)**

killed by the space restriction:

$$\left[\left[\hat{F},\,\hat{T}_2\right]^{\dagger},\,\hat{T}_2\right] = f^{i_1}_{i_2} t^{a_1 a_2}_{i_1 i_3} t^{a_3 a_4}_{i_2 i_3} \hat{E}^{a_1 a_2}_{a_3 a_4} \tag{21}$$

$$- f^{a_4}_{p_1} t^{a_1 a_2}_{i_1 i_2} t^{a_4 a_3}_{i_2 i_1} \hat{E}^{a_1 a_2}_{a_3 p_1} \tag{22}$$

$$+ f^{i_3}_{i_4} t^{a_1 a_2}_{i_3 i_2} t^{a_3 a_4}_{i_4 i_1} \hat{E}^{a_1 a_2 i_1}_{a_3 i_2 a_4} \tag{23}$$

$$- f^{a_4}_{p_1} t^{a_1 a_2}_{i_3 i_2} t^{a_4 a_3}_{i_1 i_3} \hat{E}^{a_1 a_2 i_1}_{a_3 i_2 p_1} \tag{24}$$

$$- f^{a_4}_{p_1} t^{a_1 a_2}_{i_2 i_3} t^{a_4 a_3}_{i_3 i_1} \hat{E}^{a_1 a_2 i_1}_{i_2 p_1 a_3} \tag{25}$$

$$+ f^{i_2}_{p_1} t^{a_1 a_2}_{i_3 i_1} t^{a_3 a_4}_{i_3 i_2} \hat{E}^{a_1 a_2 p_1}_{a_3 i_1 a_4} \tag{26}$$

the 4 three-body terms fold into 6 terms by the cumulant template:

$$\text{3-body part} = -\tfrac{1}{2}\, D^{i_1}_{i_2} f^{i_3}_{i_4} t^{a_1 a_2}_{i_1 i_3} t^{a_3 a_4}_{i_2 i_4} \hat{E}^{a_1 a_2}_{a_3 a_4} \tag{27}$$

$$- \tfrac{1}{2}\, D^{i_1}_{p_1} f^{i_2}_{p_1} t^{a_1 a_2}_{i_1 i_3} t^{a_3 a_4}_{i_2 i_3} \hat{E}^{a_1 a_2}_{a_3 a_4} \tag{28}$$

$$- D^{i_2}_{p_1} f^{a_4}_{p_1} t^{a_1 a_2}_{i_3 i_1} t^{a_4 a_3}_{i_2 i_3} \hat{E}^{a_1 a_2}_{a_3 i_1} \tag{29}$$

$$+ \tfrac{1}{2}\, D^{i_2}_{p_1} f^{a_4}_{p_1} t^{a_1 a_2}_{i_3 i_1} t^{a_4 a_3}_{i_3 i_2} \hat{E}^{a_1 a_2}_{a_3 i_1} \tag{30}$$

$$+ \tfrac{1}{2}\, D^{i_1}_{i_2} f^{a_4}_{p_1} t^{a_1 a_2}_{i_1 i_3} t^{a_4 a_3}_{i_3 i_2} \hat{E}^{a_1 a_2}_{a_3 p_1} \tag{31}$$

$$+ \tfrac{1}{2}\, D^{i_1}_{i_2} f^{a_4}_{p_1} t^{a_1 a_2}_{i_3 i_1} t^{a_4 a_3}_{i_2 i_3} \hat{E}^{a_1 a_2}_{a_3 p_1} \tag{32}$$

the special block (killed terms and folded pieces merged, 8 terms):

$$\left[\left[\hat{F},\,\hat{T}_2\right]^{\dagger},\,\hat{T}_2\right]^{F12} = -\tfrac{1}{2}\, D^{i_1}_{i_2} f^{i_3}_{i_4} t^{a_1 a_2}_{i_1 i_3} t^{a_3 a_4}_{i_2 i_4} \hat{E}^{a_1 a_2}_{a_3 a_4} \tag{33}$$

$$- \tfrac{1}{2}\, D^{i_1}_{p_1} f^{i_2}_{p_1} t^{a_1 a_2}_{i_1 i_3} t^{a_3 a_4}_{i_2 i_3} \hat{E}^{a_1 a_2}_{a_3 a_4} \tag{34}$$

$$+ f^{i_1}_{i_2} t^{a_1 a_2}_{i_1 i_3} t^{a_3 a_4}_{i_2 i_3} \hat{E}^{a_1 a_2}_{a_3 a_4} \tag{35}$$

$$- D^{i_2}_{p_1} f^{a_4}_{p_1} t^{a_1 a_2}_{i_3 i_1} t^{a_4 a_3}_{i_2 i_3} \hat{E}^{a_1 a_2}_{a_3 i_1} \tag{36}$$

$$+ \tfrac{1}{2}\, D^{i_2}_{p_1} f^{a_4}_{p_1} t^{a_1 a_2}_{i_3 i_1} t^{a_4 a_3}_{i_3 i_2} \hat{E}^{a_1 a_2}_{a_3 i_1} \tag{37}$$

$$+ \tfrac{1}{2}\, D^{i_1}_{i_2} f^{a_4}_{p_1} t^{a_1 a_2}_{i_1 i_3} t^{a_4 a_3}_{i_3 i_2} \hat{E}^{a_1 a_2}_{a_3 p_1} \tag{38}$$

$$+ \tfrac{1}{2}\, D^{i_1}_{i_2} f^{a_4}_{p_1} t^{a_1 a_2}_{i_3 i_1} t^{a_4 a_3}_{i_2 i_3} \hat{E}^{a_1 a_2}_{a_3 p_1} \tag{39}$$

$$- f^{a_4}_{p_1} t^{a_1 a_2}_{i_1 i_2} t^{a_4 a_3}_{i_2 i_1} \hat{E}^{a_1 a_2}_{a_3 p_1} \tag{40}$$

2 The assembled F12 special term

Assembled the way the commutator stage defines the transcorrelated Hamiltonian: the blocks plus their Hermitian conjugates (the anti-Hermitian generator $\hat{A} = \hat{T} - \hat{T}^\dagger$), the second-order Fock blocks carrying the factor 1/2, merged once (32 terms):

$$
\begin{aligned}
\bar{H}^{F12}_{\text{special}} = & -\tfrac{1}{2}\, D^{i_1}_{i_2} f^{i_3}_{i_4} t^{a_1 a_2}_{i_1 i_3} t^{a_3 a_4}_{i_2 i_4} \hat{E}^{a_1 a_2}_{a_3 a_4} && (41)\\
& -\tfrac{1}{4}\, D^{i_1}_{p_1} f^{i_2}_{p_1} t^{a_1 a_2}_{i_1 i_3} t^{a_3 a_4}_{i_2 i_3} \hat{E}^{a_1 a_2}_{a_3 a_4} && (42)\\
& -\tfrac{1}{4}\, D^{i_1}_{p_1} f^{i_2}_{p_1} t^{a_1 a_2}_{i_2 i_3} t^{a_3 a_4}_{i_1 i_3} \hat{E}^{a_1 a_2}_{a_3 a_4} && (43)\\
& +\, f^{i_1}_{i_2} t^{a_1 a_2}_{i_1 i_3} t^{a_3 a_4}_{i_2 i_3} \hat{E}^{a_1 a_2}_{a_3 a_4} && (44)\\
& -\tfrac{1}{2}\, D^{i_2}_{p_1} f^{a_4}_{p_1} t^{a_1 a_2}_{i_3 i_1} t^{a_4 a_3}_{i_2 i_3} \hat{E}^{a_1 a_2}_{a_3 i_1} && (45)\\
& +\tfrac{1}{4}\, D^{i_2}_{p_1} f^{a_4}_{p_1} t^{a_1 a_2}_{i_3 i_1} t^{a_4 a_3}_{i_3 i_2} \hat{E}^{a_1 a_2}_{a_3 i_1} && (46)\\
& +\, f^{i_2}_{i_3} t^{a_1 a_2}_{i_2 i_1} t^{a_3}_{i_3} \hat{E}^{a_1 a_2}_{a_3 i_1} && (47)\\
& +\tfrac{1}{4}\, D^{i_1}_{i_2} f^{a_4}_{p_1} t^{a_1 a_2}_{i_1 i_3} t^{a_4 a_3}_{i_3 i_2} \hat{E}^{a_1 a_2}_{a_3 p_1} && (48)\\
& +\tfrac{1}{4}\, D^{i_1}_{i_2} f^{a_4}_{p_1} t^{a_1 a_2}_{i_3 i_1} t^{a_4 a_3}_{i_2 i_3} \hat{E}^{a_1 a_2}_{a_3 p_1} && (49)\\
& -\tfrac{1}{2}\, f^{a_4}_{p_1} t^{a_1 a_2}_{i_1 i_2} t^{a_4 a_3}_{i_2 i_1} \hat{E}^{a_1 a_2}_{a_3 p_1} && (50)\\
& +\tfrac{1}{2}\, f^{i_1}_{p_1} t^{a_1 a_2}_{i_2 i_1} t^{a_3}_{i_2} \hat{E}^{a_1 a_2}_{a_3 p_1} && (51)\\
& -\, f^{a_3}_{i_3} t^{a_1 a_2}_{i_1 i_3} t^{a_3}_{i_2} \hat{E}^{a_1 a_2}_{i_1 i_2} && (52)\\
& -\, D^{p_2}_{p_3} g^{i_2 p_2}_{p_1 p_3} t^{a_1 a_2}_{i_1 i_2} \hat{E}^{a_1 a_2}_{i_1 p_1} && (53)\\
& +\tfrac{1}{2}\, D^{p_2}_{p_3} g^{i_2 p_2}_{p_3 p_1} t^{a_1 a_2}_{i_1 i_2} \hat{E}^{a_1 a_2}_{i_1 p_1} && (54)\\
& -\tfrac{1}{2}\, f^{a_3}_{p_1} t^{a_1 a_2}_{i_1 i_2} t^{a_3}_{i_2} \hat{E}^{a_1 a_2}_{i_1 p_1} && (55)\\
& -\, h^{i_2}_{p_1} t^{a_1 a_2}_{i_1 i_2} \hat{E}^{a_1 a_2}_{i_1 p_1} && (56)\\
& +\tfrac{1}{2}\, D^{i_1}_{p_3} g^{i_2 p_3}_{p_1 p_2} t^{a_1 a_2}_{i_2 i_1} \hat{E}^{a_1 a_2}_{p_1 p_2} && (57)\\
& -\tfrac{1}{2}\, g^{i_1 i_2}_{p_1 p_2} t^{a_1 a_2}_{i_1 i_2} \hat{E}^{a_1 a_2}_{p_1 p_2} && (58)\\
& -\tfrac{1}{2}\, D^{i_2}_{p_1} f^{a_4}_{p_1} t^{a_2 a_3}_{i_3 i_1} t^{a_4 a_1}_{i_2 i_3} \hat{E}^{a_1 i_1}_{a_2 a_3} && (59)\\
& +\tfrac{1}{4}\, D^{i_2}_{p_1} f^{a_4}_{p_1} t^{a_2 a_3}_{i_3 i_1} t^{a_4 a_1}_{i_3 i_2} \hat{E}^{a_1 i_1}_{a_2 a_3} && (60)\\
& +\, f^{i_2}_{i_3} t^{a_1}_{i_2} t^{a_2 a_3}_{i_3 i_1} \hat{E}^{a_1 i_1}_{a_2 a_3} && (61)\\
& +\tfrac{1}{4}\, D^{i_1}_{i_2} f^{a_4}_{p_1} t^{a_2 a_3}_{i_1 i_3} t^{a_4 a_1}_{i_3 i_2} \hat{E}^{a_1 p_1}_{a_2 a_3} && (62)\\
& +\tfrac{1}{4}\, D^{i_1}_{i_2} f^{a_4}_{p_1} t^{a_2 a_3}_{i_3 i_1} t^{a_4 a_1}_{i_2 i_3} \hat{E}^{a_1 p_1}_{a_2 a_3} && (63)\\
& -\tfrac{1}{2}\, f^{a_4}_{p_1} t^{a_2 a_3}_{i_1 i_2} t^{a_4 a_1}_{i_2 i_1} \hat{E}^{a_1 p_1}_{a_2 a_3} && (64)\\
& +\tfrac{1}{2}\, f^{i_1}_{p_1} t^{a_1}_{i_2} t^{a_2 a_3}_{i_2 i_1} \hat{E}^{a_1 p_1}_{a_2 a_3} && (65)\\
& -\, f^{a_3}_{i_3} t^{a_1 a_2}_{i_1 i_3} t^{a_3}_{i_2} \hat{E}^{i_1 i_2}_{a_1 a_2} && (66)\\
& -\, D^{p_2}_{p_3} g^{i_2 p_2}_{p_1 p_3} t^{a_1 a_2}_{i_1 i_2} \hat{E}^{i_1 p_1}_{a_1 a_2} && (67)\\
& +\tfrac{1}{2}\, D^{p_2}_{p_3} g^{i_2 p_2}_{p_3 p_1} t^{a_1 a_2}_{i_1 i_2} \hat{E}^{i_1 p_1}_{a_1 a_2} && (68)\\
& -\tfrac{1}{2}\, f^{a_3}_{p_1} t^{a_1 a_2}_{i_1 i_2} t^{a_3}_{i_2} \hat{E}^{i_1 p_1}_{a_1 a_2} && (69)\\
& -\, h^{i_2}_{p_1} t^{a_1 a_2}_{i_1 i_2} \hat{E}^{i_1 p_1}_{a_1 a_2} && (70)\\
& +\tfrac{1}{2}\, D^{i_1}_{p_3} g^{i_2 p_3}_{p_1 p_2} t^{a_1 a_2}_{i_2 i_1} \hat{E}^{p_1 p_2}_{a_1 a_2} && (71)
\end{aligned}
$$

$$- \tfrac{1}{2}\, g^{i_1 i_2}_{p_1 p_2}\, t^{a_1 a_2}_{i_1 i_2}\, \hat{E}^{p_1 p_2}_{a_1 a_2} \tag{72}$$

Every term carries a geminal amplitude with both virtual indices on the operator, so the whole expression vanishes when the free virtuals are OBS-vir (the SP geminal has no both-OBS-vir amplitudes) – substituting it back does not touch the restricted result, eq. (S1) of the reference. The pure-T_2 part (the doubles contribution of the geminal operator $\hat{A}^{F12}$, 24 terms):

$$\begin{aligned}
\bar{H}^{F12}_{\text{special}}\Big|_{\text{doubles}} = &-\tfrac{1}{2}\, D^{i_1}_{i_2}\, f^{i_3}_{i_4}\, t^{a_1 a_2}_{i_1 i_3}\, t^{a_3 a_4}_{i_2 i_4}\, \hat{E}^{a_1 a_2}_{a_3 a_4} && (73)\\
&- \tfrac{1}{4}\, D^{i_1}_{p_1}\, f^{i_2}_{p_1}\, t^{a_1 a_2}_{i_1 i_3}\, t^{a_3 a_4}_{i_2 i_3}\, \hat{E}^{a_1 a_2}_{a_3 a_4} && (74)\\
&- \tfrac{1}{4}\, D^{i_1}_{p_1}\, f^{i_2}_{p_1}\, t^{a_1 a_2}_{i_2 i_3}\, t^{a_3 a_4}_{i_1 i_3}\, \hat{E}^{a_1 a_2}_{a_3 a_4} && (75)\\
&+ f^{i_1}_{i_2}\, t^{a_1 a_2}_{i_1 i_3}\, t^{a_3 a_4}_{i_2 i_3}\, \hat{E}^{a_1 a_2}_{a_3 a_4} && (76)\\
&- \tfrac{1}{2}\, D^{i_2}_{p_1}\, f^{a_4}_{p_1}\, t^{a_1 a_2}_{i_3 i_1}\, t^{a_4 a_3}_{i_2 i_3}\, \hat{E}^{a_1 a_2}_{a_3 i_1} && (77)\\
&+ \tfrac{1}{4}\, D^{i_2}_{p_1}\, f^{a_4}_{p_1}\, t^{a_1 a_2}_{i_3 i_1}\, t^{a_4 a_3}_{i_3 i_2}\, \hat{E}^{a_1 a_2}_{a_3 i_1} && (78)\\
&+ \tfrac{1}{4}\, D^{i_1}_{i_2}\, f^{a_4}_{p_1}\, t^{a_1 a_2}_{i_1 i_3}\, t^{a_4 a_3}_{i_3 i_2}\, \hat{E}^{a_1 a_2}_{a_3 p_1} && (79)\\
&+ \tfrac{1}{4}\, D^{i_1}_{i_2}\, f^{a_4}_{p_1}\, t^{a_1 a_2}_{i_3 i_1}\, t^{a_4 a_3}_{i_2 i_3}\, \hat{E}^{a_1 a_2}_{a_3 p_1} && (80)\\
&- \tfrac{1}{2}\, f^{a_4}_{p_1}\, t^{a_1 a_2}_{i_1 i_2}\, t^{a_4 a_3}_{i_2 i_1}\, \hat{E}^{a_1 a_2}_{a_3 p_1} && (81)\\
&- D^{p_2}_{p_3}\, g^{i_2 p_2}_{p_1 p_3}\, t^{a_1 a_2}_{i_1 i_2}\, \hat{E}^{a_1 a_2}_{i_1 p_1} && (82)\\
&+ \tfrac{1}{2}\, D^{p_2}_{p_3}\, g^{i_2 p_2}_{p_3 p_1}\, t^{a_1 a_2}_{i_1 i_2}\, \hat{E}^{a_1 a_2}_{i_1 p_1} && (83)\\
&- h^{i_2}_{p_1}\, t^{a_1 a_2}_{i_1 i_2}\, \hat{E}^{a_1 a_2}_{i_1 p_1} && (84)\\
&+ \tfrac{1}{2}\, D^{i_1}_{p_3}\, g^{i_2 p_3}_{p_1 p_2}\, t^{a_1 a_2}_{i_2 i_1}\, \hat{E}^{a_1 a_2}_{p_1 p_2} && (85)\\
&- \tfrac{1}{2}\, g^{i_1 i_2}_{p_1 p_2}\, t^{a_1 a_2}_{i_1 i_2}\, \hat{E}^{a_1 a_2}_{p_1 p_2} && (86)\\
&- \tfrac{1}{2}\, D^{i_2}_{p_1}\, f^{a_4}_{p_1}\, t^{a_2 a_3}_{i_3 i_1}\, t^{a_4 a_1}_{i_2 i_3}\, \hat{E}^{a_1 i_1}_{a_2 a_3} && (87)\\
&+ \tfrac{1}{4}\, D^{i_2}_{p_1}\, f^{a_4}_{p_1}\, t^{a_2 a_3}_{i_3 i_1}\, t^{a_4 a_1}_{i_3 i_2}\, \hat{E}^{a_1 i_1}_{a_2 a_3} && (88)\\
&+ \tfrac{1}{4}\, D^{i_1}_{i_2}\, f^{a_4}_{p_1}\, t^{a_2 a_3}_{i_1 i_3}\, t^{a_4 a_1}_{i_3 i_2}\, \hat{E}^{a_1 p_1}_{a_2 a_3} && (89)\\
&+ \tfrac{1}{4}\, D^{i_1}_{i_2}\, f^{a_4}_{p_1}\, t^{a_2 a_3}_{i_3 i_1}\, t^{a_4 a_1}_{i_2 i_3}\, \hat{E}^{a_1 p_1}_{a_2 a_3} && (90)\\
&- \tfrac{1}{2}\, f^{a_4}_{p_1}\, t^{a_2 a_3}_{i_1 i_2}\, t^{a_4 a_1}_{i_2 i_1}\, \hat{E}^{a_1 p_1}_{a_2 a_3} && (91)\\
&- D^{p_2}_{p_3}\, g^{i_2 p_2}_{p_1 p_3}\, t^{a_1 a_2}_{i_1 i_2}\, \hat{E}^{i_1 p_1}_{a_1 a_2} && (92)\\
&+ \tfrac{1}{2}\, D^{p_2}_{p_3}\, g^{i_2 p_2}_{p_3 p_1}\, t^{a_1 a_2}_{i_1 i_2}\, \hat{E}^{i_1 p_1}_{a_1 a_2} && (93)\\
&- h^{i_2}_{p_1}\, t^{a_1 a_2}_{i_1 i_2}\, \hat{E}^{i_1 p_1}_{a_1 a_2} && (94)\\
&+ \tfrac{1}{2}\, D^{i_1}_{p_3}\, g^{i_2 p_3}_{p_1 p_2}\, t^{a_1 a_2}_{i_2 i_1}\, \hat{E}^{p_1 p_2}_{a_1 a_2} && (95)\\
&- \tfrac{1}{2}\, g^{i_1 i_2}_{p_1 p_2}\, t^{a_1 a_2}_{i_1 i_2}\, \hat{E}^{p_1 p_2}_{a_1 a_2} && (96)
\end{aligned}$$

3 The substituted final formula

The stage-6 result with the F12 special term substituted back, assembled (blocks + h.c., 1/2 on the Fock double commutator; 166 terms) – the complete transcorrelated Hamiltonian through $O(\lambda^2)$ with the cumulant-folded three-body part and the fixed geminal amplitudes:

$$\bar{H} - \hat{H} = -\tfrac{1}{2}\, D^{i_1 i_2}_{i_3 i_4}\, f^{a_3}_{a_4}\, t^{a_3 a_1}_{i_1 i_2}\, t^{a_4 a_2}_{i_3 i_4}\, \hat{E}^{a_1}_{a_2} \tag{97}$$

$$+ \frac{1}{4} D^{i_1 i_2}_{i_3 p_1} f^{i_4}_{p_1} t^{a_3 a_1}_{i_1 i_2} t^{a_3 a_2}_{i_3 i_4} \hat{E}^{a_1}_{a_2} \tag{98}$$
$$+ \frac{1}{4} D^{i_1 i_2}_{i_3 p_1} f^{i_4}_{p_1} t^{a_3 a_1}_{i_2 i_1} t^{a_3 a_2}_{i_4 i_3} \hat{E}^{a_1}_{a_2} \tag{99}$$
$$+ \frac{1}{4} D^{i_1 i_2}_{i_3 p_1} f^{i_4}_{p_1} t^{a_3 a_1}_{i_3 i_4} t^{a_3 a_2}_{i_1 i_2} \hat{E}^{a_1}_{a_2} \tag{100}$$
$$+ \frac{1}{4} D^{i_1 i_2}_{i_3 p_1} f^{i_4}_{p_1} t^{a_3 a_1}_{i_4 i_3} t^{a_3 a_2}_{i_2 i_1} \hat{E}^{a_1}_{a_2} \tag{101}$$
$$+ D^{i_1}_{i_2} D^{i_3}_{i_4} f^{a_3}_{a_4} t^{a_3 a_1}_{i_1 i_3} t^{a_4 a_2}_{i_2 i_4} \hat{E}^{a_1}_{a_2} \tag{102}$$
$$- \frac{1}{2} D^{i_1}_{i_2} D^{i_3}_{i_4} f^{a_3}_{a_4} t^{a_3 a_1}_{i_1 i_3} t^{a_4 a_2}_{i_4 i_2} \hat{E}^{a_1}_{a_2} \tag{103}$$
$$- \frac{1}{2} D^{i_1}_{i_2} D^{i_3}_{p_1} f^{i_4}_{p_1} t^{a_3 a_1}_{i_1 i_3} t^{a_3 a_2}_{i_2 i_4} \hat{E}^{a_1}_{a_2} \tag{104}$$
$$+ \frac{1}{4} D^{i_1}_{i_2} D^{i_3}_{p_1} f^{i_4}_{p_1} t^{a_3 a_1}_{i_1 i_3} t^{a_3 a_2}_{i_4 i_2} \hat{E}^{a_1}_{a_2} \tag{105}$$
$$- \frac{1}{2} D^{i_1}_{i_2} D^{i_3}_{p_1} f^{i_4}_{p_1} t^{a_3 a_1}_{i_1 i_4} t^{a_3 a_2}_{i_2 i_3} \hat{E}^{a_1}_{a_2} \tag{106}$$
$$+ \frac{1}{4} D^{i_1}_{i_2} D^{i_3}_{p_1} f^{i_4}_{p_1} t^{a_3 a_1}_{i_1 i_4} t^{a_3 a_2}_{i_3 i_2} \hat{E}^{a_1}_{a_2} \tag{107}$$
$$+ \frac{1}{4} D^{i_1}_{i_2} D^{i_3}_{p_1} f^{i_4}_{p_1} t^{a_3 a_1}_{i_3 i_1} t^{a_3 a_2}_{i_2 i_4} \hat{E}^{a_1}_{a_2} \tag{108}$$
$$- \frac{1}{2} D^{i_1}_{i_2} D^{i_3}_{p_1} f^{i_4}_{p_1} t^{a_3 a_1}_{i_3 i_1} t^{a_3 a_2}_{i_4 i_2} \hat{E}^{a_1}_{a_2} \tag{109}$$
$$+ \frac{1}{4} D^{i_1}_{i_2} D^{i_3}_{p_1} f^{i_4}_{p_1} t^{a_3 a_1}_{i_4 i_1} t^{a_3 a_2}_{i_2 i_3} \hat{E}^{a_1}_{a_2} \tag{110}$$
$$- \frac{1}{2} D^{i_1}_{i_2} D^{i_3}_{p_1} f^{i_4}_{p_1} t^{a_3 a_1}_{i_4 i_1} t^{a_3 a_2}_{i_3 i_2} \hat{E}^{a_1}_{a_2} \tag{111}$$
$$+ f^{i_1}_{i_2} t^{a_1}_{i_1} t^{a_2}_{i_2} \hat{E}^{a_1}_{a_2} \tag{112}$$
$$- \frac{1}{4} D^{i_2 i_3}_{i_4 p_1} f^{a_2}_{p_1} t^{a_2 a_3}_{i_3 i_2} t^{a_3 a_1}_{i_1 i_4} \hat{E}^{a_1}_{i_1} \tag{113}$$
$$+ \frac{1}{2} D^{i_2 i_3}_{i_4 p_1} f^{a_2}_{p_1} t^{a_2 a_3}_{i_3 i_2} t^{a_3 a_1}_{i_4 i_1} \hat{E}^{a_1}_{i_1} \tag{114}$$
$$- \frac{1}{2} D^{i_2 p_1}_{p_2 p_3} g^{a_2 p_1}_{p_2 p_3} t^{a_2 a_1}_{i_1 i_2} \hat{E}^{a_1}_{i_1} \tag{115}$$
$$+ D^{i_2 p_1}_{p_2 p_3} g^{a_2 p_1}_{p_2 p_3} t^{a_2 a_1}_{i_2 i_1} \hat{E}^{a_1}_{i_1} \tag{116}$$
$$- \frac{1}{4} D^{i_2}_{i_3} D^{i_4}_{p_1} f^{a_2}_{p_1} t^{a_2 a_3}_{i_2 i_4} t^{a_3 a_1}_{i_1 i_3} \hat{E}^{a_1}_{i_1} \tag{117}$$
$$+ \frac{1}{2} D^{i_2}_{i_3} D^{i_4}_{p_1} f^{a_2}_{p_1} t^{a_2 a_3}_{i_2 i_4} t^{a_3 a_1}_{i_3 i_1} \hat{E}^{a_1}_{i_1} \tag{118}$$
$$+ \frac{1}{2} D^{i_2}_{i_3} D^{i_4}_{p_1} f^{a_2}_{p_1} t^{a_2 a_3}_{i_4 i_2} t^{a_3 a_1}_{i_1 i_3} \hat{E}^{a_1}_{i_1} \tag{119}$$
$$- D^{i_2}_{i_3} D^{i_4}_{p_1} f^{a_2}_{p_1} t^{a_2 a_3}_{i_4 i_2} t^{a_3 a_1}_{i_3 i_1} \hat{E}^{a_1}_{i_1} \tag{120}$$
$$+ D^{i_2}_{p_1} D^{p_2}_{p_3} g^{a_2 p_2}_{p_1 p_3} t^{a_2 a_1}_{i_1 i_2} \hat{E}^{a_1}_{i_1} \tag{121}$$
$$- 2 D^{i_2}_{p_1} D^{p_2}_{p_3} g^{a_2 p_2}_{p_1 p_3} t^{a_2 a_1}_{i_2 i_1} \hat{E}^{a_1}_{i_1} \tag{122}$$
$$- \frac{1}{2} D^{i_2}_{p_1} D^{p_2}_{p_3} g^{a_2 p_2}_{p_3 p_1} t^{a_2 a_1}_{i_1 i_2} \hat{E}^{a_1}_{i_1} \tag{123}$$
$$+ D^{i_2}_{p_1} D^{p_2}_{p_3} g^{a_2 p_2}_{p_3 p_1} t^{a_2 a_1}_{i_2 i_1} \hat{E}^{a_1}_{i_1} \tag{124}$$
$$- f^{a_2}_{i_2} t^{a_1}_{i_2} t^{a_2}_{i_1} \hat{E}^{a_1}_{i_1} \tag{125}$$
$$- \frac{1}{4} D^{i_1 i_2}_{i_3 i_4} f^{a_2}_{p_1} t^{a_2 a_3}_{i_1 i_2} t^{a_3 a_1}_{i_4 i_3} \hat{E}^{a_1}_{p_1} \tag{126}$$
$$- \frac{1}{2} D^{i_1 i_2}_{p_2 p_3} g^{a_2 p_1}_{p_2 p_3} t^{a_2 a_1}_{i_1 i_2} \hat{E}^{a_1}_{p_1} \tag{127}$$
$$- \frac{1}{4} D^{i_1}_{i_2} D^{i_3}_{i_4} f^{a_2}_{p_1} t^{a_2 a_3}_{i_1 i_3} t^{a_3 a_1}_{i_2 i_4} \hat{E}^{a_1}_{p_1} \tag{128}$$
$$+ \frac{1}{2} D^{i_1}_{i_2} D^{i_3}_{i_4} f^{a_2}_{p_1} t^{a_2 a_3}_{i_1 i_3} t^{a_3 a_1}_{i_4 i_2} \hat{E}^{a_1}_{p_1} \tag{129}$$
$$+ D^{i_1}_{p_2} D^{i_2}_{p_3} g^{a_2 p_1}_{p_2 p_3} t^{a_2 a_1}_{i_1 i_2} \hat{E}^{a_1}_{p_1} \tag{130}$$
$$- \frac{1}{2} D^{i_1}_{p_2} D^{i_2}_{p_3} g^{a_2 p_1}_{p_2 p_3} t^{a_2 a_1}_{i_2 i_1} \hat{E}^{a_1}_{p_1} \tag{131}$$
$$- \frac{1}{2} f^{a_2}_{p_1} t^{a_1}_{i_1} t^{a_2}_{i_1} \hat{E}^{a_1}_{p_1} - h^{i_1}_{p_1} t^{a_1}_{i_1} \hat{E}^{a_1}_{p_1} \tag{132}$$

$$- \frac{1}{4} D^{i_2 i_3}_{i_4 p_1} f^{a_2}_{p_1} t^{a_2 a_3}_{i_3 i_2} t^{a_3 a_1}_{i_1 i_4} \hat{E}^{i_1}_{a_1} \tag{133}$$

$$+ \frac{1}{2} D^{i_2 i_3}_{i_4 p_1} f^{a_2}_{p_1} t^{a_2 a_3}_{i_3 i_2} t^{a_3 a_1}_{i_4 i_1} \hat{E}^{i_1}_{a_1} \tag{134}$$

$$- \frac{1}{2} D^{i_2 p_1}_{p_2 p_3} g^{a_2 p_1}_{p_2 p_3} t^{a_2 a_1}_{i_1 i_2} \hat{E}^{i_1}_{a_1} \tag{135}$$

$$+ D^{i_2 p_1}_{p_2 p_3} g^{a_2 p_1}_{p_2 p_3} t^{a_2 a_1}_{i_2 i_1} \hat{E}^{i_1}_{a_1} \tag{136}$$

$$- \frac{1}{4} D^{i_2}_{i_3} D^{i_4}_{p_1} f^{a_2}_{p_1} t^{a_2 a_3}_{i_2 i_4} t^{a_3 a_1}_{i_1 i_3} \hat{E}^{i_1}_{a_1} \tag{137}$$

$$+ \frac{1}{2} D^{i_2}_{i_3} D^{i_4}_{p_1} f^{a_2}_{p_1} t^{a_2 a_3}_{i_2 i_4} t^{a_3 a_1}_{i_3 i_1} \hat{E}^{i_1}_{a_1} \tag{138}$$

$$+ \frac{1}{2} D^{i_2}_{i_3} D^{i_4}_{p_1} f^{a_2}_{p_1} t^{a_2 a_3}_{i_4 i_2} t^{a_3 a_1}_{i_1 i_3} \hat{E}^{i_1}_{a_1} \tag{139}$$

$$- D^{i_2}_{i_3} D^{i_4}_{p_1} f^{a_2}_{p_1} t^{a_2 a_3}_{i_4 i_2} t^{a_3 a_1}_{i_3 i_1} \hat{E}^{i_1}_{a_1} \tag{140}$$

$$+ D^{i_2}_{p_1} D^{p_2}_{p_3} g^{a_2 p_2}_{p_1 p_3} t^{a_2 a_1}_{i_1 i_2} \hat{E}^{i_1}_{a_1} \tag{141}$$

$$- 2 D^{i_2}_{p_1} D^{p_2}_{p_3} g^{a_2 p_2}_{p_1 p_3} t^{a_2 a_1}_{i_2 i_1} \hat{E}^{i_1}_{a_1} \tag{142}$$

$$- \frac{1}{2} D^{i_2}_{p_1} D^{p_2}_{p_3} g^{a_2 p_2}_{p_3 p_1} t^{a_2 a_1}_{i_1 i_2} \hat{E}^{i_1}_{a_1} \tag{143}$$

$$+ D^{i_2}_{p_1} D^{p_2}_{p_3} g^{a_2 p_2}_{p_3 p_1} t^{a_2 a_1}_{i_2 i_1} \hat{E}^{i_1}_{a_1} \tag{144}$$

$$- f^{a_2}_{i_2} t^{a_1}_{i_2} t^{a_2}_{i_1} \hat{E}^{i_1}_{a_1} + f^{a_1}_{a_2} t^{a_1}_{i_1} t^{a_2}_{i_2} \hat{E}^{i_1}_{i_2} \tag{145}$$

$$- \frac{1}{2} f^{i_2}_{p_1} t^{a_1}_{i_1} t^{a_1}_{i_2} \hat{E}^{i_1}_{p_1} + h^{a_1}_{p_1} t^{a_1}_{i_1} \hat{E}^{i_1}_{p_1} \tag{146}$$

$$- \frac{1}{4} D^{i_1 i_2}_{i_3 i_4} f^{a_2}_{p_1} t^{a_2 a_3}_{i_1 i_2} t^{a_3 a_1}_{i_4 i_3} \hat{E}^{p_1}_{a_1} \tag{147}$$

$$- \frac{1}{2} D^{i_1 i_2}_{p_2 p_3} g^{a_2 p_1}_{p_2 p_3} t^{a_2 a_1}_{i_1 i_2} \hat{E}^{p_1}_{a_1} \tag{148}$$

$$- \frac{1}{4} D^{i_1}_{i_2} D^{i_3}_{i_4} f^{a_2}_{p_1} t^{a_2 a_3}_{i_1 i_3} t^{a_3 a_1}_{i_2 i_4} \hat{E}^{p_1}_{a_1} \tag{149}$$

$$+ \frac{1}{2} D^{i_1}_{i_2} D^{i_3}_{i_4} f^{a_2}_{p_1} t^{a_2 a_3}_{i_1 i_3} t^{a_3 a_1}_{i_4 i_2} \hat{E}^{p_1}_{a_1} \tag{150}$$

$$+ D^{i_1}_{p_2} D^{i_2}_{p_3} g^{a_2 p_1}_{p_2 p_3} t^{a_2 a_1}_{i_1 i_2} \hat{E}^{p_1}_{a_1} \tag{151}$$

$$- \frac{1}{2} D^{i_1}_{p_2} D^{i_2}_{p_3} g^{a_2 p_1}_{p_2 p_3} t^{a_2 a_1}_{i_2 i_1} \hat{E}^{p_1}_{a_1} \tag{152}$$

$$- \frac{1}{2} f^{a_2}_{p_1} t^{a_1}_{i_1} t^{a_2}_{i_1} \hat{E}^{p_1}_{a_1} - h^{i_1}_{p_1} t^{a_1}_{i_1} \hat{E}^{p_1}_{a_1} \tag{153}$$

$$- \frac{1}{2} f^{i_2}_{p_1} t^{a_1}_{i_1} t^{a_1}_{i_2} \hat{E}^{p_1}_{i_1} + h^{a_1}_{p_1} t^{a_1}_{i_1} \hat{E}^{p_1}_{i_1} \tag{154}$$

$$- \frac{1}{2} D^{i_1}_{i_2} f^{i_3}_{i_4} t^{a_1 a_2}_{i_1 i_3} t^{a_3 a_4}_{i_2 i_4} \hat{E}^{a_1 a_2}_{a_3 a_4} \tag{155}$$

$$- \frac{1}{4} D^{i_1}_{p_1} f^{i_2}_{p_1} t^{a_1 a_2}_{i_1 i_3} t^{a_3 a_4}_{i_2 i_3} \hat{E}^{a_1 a_2}_{a_3 a_4} \tag{156}$$

$$- \frac{1}{4} D^{i_1}_{p_1} f^{i_2}_{p_1} t^{a_1 a_2}_{i_2 i_3} t^{a_3 a_4}_{i_1 i_3} \hat{E}^{a_1 a_2}_{a_3 a_4} \tag{157}$$

$$+ f^{i_1}_{i_2} t^{a_1 a_2}_{i_1 i_3} t^{a_3 a_4}_{i_2 i_3} \hat{E}^{a_1 a_2}_{a_3 a_4} \tag{158}$$

$$- \frac{1}{2} D^{i_2}_{p_1} f^{a_4}_{p_1} t^{a_1 a_2}_{i_3 i_1} t^{a_4 a_3}_{i_2 i_3} \hat{E}^{a_1 a_2}_{a_3 i_1} \tag{159}$$

$$+ \frac{1}{4} D^{i_2}_{p_1} f^{a_4}_{p_1} t^{a_1 a_2}_{i_3 i_1} t^{a_4 a_3}_{i_3 i_2} \hat{E}^{a_1 a_2}_{a_3 i_1} \tag{160}$$

$$+ f^{i_2}_{i_3} t^{a_1 a_2}_{i_2 i_1} t^{a_3}_{i_3} \hat{E}^{a_1 a_2}_{a_3 i_1} \tag{161}$$

$$+ \frac{1}{4} D^{i_1}_{i_2} f^{a_4}_{p_1} t^{a_1 a_2}_{i_1 i_3} t^{a_4 a_3}_{i_3 i_2} \hat{E}^{a_1 a_2}_{a_3 p_1} \tag{162}$$

$$+ \frac{1}{4} D^{i_1}_{i_2} f^{a_4}_{p_1} t^{a_1 a_2}_{i_3 i_1} t^{a_4 a_3}_{i_2 i_3} \hat{E}^{a_1 a_2}_{a_3 p_1} \tag{163}$$

$$- \frac{1}{2} f^{a_4}_{p_1} t^{a_1 a_2}_{i_1 i_2} t^{a_4 a_3}_{i_2 i_1} \hat{E}^{a_1 a_2}_{a_3 p_1} \tag{164}$$

$$+ \frac{1}{2} f^{i_1}_{p_1} t^{a_1 a_2}_{i_2 i_1} t^{a_3}_{i_2} \hat{E}^{a_1 a_2}_{a_3 p_1} \tag{165}$$

$$- f^{a_3}_{i_3} t^{a_1 a_2}_{i_1 i_3} t^{a_3}_{i_2} \hat{E}^{a_1 a_2}_{i_1 i_2} \tag{166}$$

$$- f^{a_3}_{i_3} t^{a_1}_{i_3} t^{a_3 a_2}_{i_1 i_2} \hat{E}^{a_1 a_2}_{i_1 i_2} \tag{167}$$

$$- D^{p_2}_{p_3} g^{i_2 p_2}_{p_1 p_3} t^{a_1 a_2}_{i_1 i_2} \hat{E}^{a_1 a_2}_{i_1 p_1} \tag{168}$$
$$+ \tfrac{1}{2} D^{p_2}_{p_3} g^{i_2 p_2}_{p_3 p_1} t^{a_1 a_2}_{i_1 i_2} \hat{E}^{a_1 a_2}_{i_1 p_1} \tag{169}$$
$$- \tfrac{1}{2} f^{a_3}_{p_1} t^{a_1 a_2}_{i_1 i_2} t^{a_3}_{i_2} \hat{E}^{a_1 a_2}_{i_1 p_1} \tag{170}$$
$$- h^{i_2}_{p_1} t^{a_1 a_2}_{i_1 i_2} \hat{E}^{a_1 a_2}_{i_1 p_1} \tag{171}$$
$$+ \tfrac{1}{2} D^{i_1}_{p_3} g^{i_2 p_3}_{p_1 p_2} t^{a_1 a_2}_{i_2 i_1} \hat{E}^{a_1 a_2}_{p_1 p_2} \tag{172}$$
$$- \tfrac{1}{2} g^{i_1 i_2}_{p_1 p_2} t^{a_1 a_2}_{i_1 i_2} \hat{E}^{a_1 a_2}_{p_1 p_2} \tag{173}$$
$$- \tfrac{1}{2} D^{i_2}_{p_1} f^{a_4}_{p_1} t^{a_2 a_3}_{i_3 i_1} t^{a_4 a_1}_{i_2 i_3} \hat{E}^{a_1 i_1}_{a_2 a_3} \tag{174}$$
$$+ \tfrac{1}{4} D^{i_2}_{p_1} f^{a_4}_{p_1} t^{a_2 a_3}_{i_3 i_1} t^{a_4 a_1}_{i_3 i_2} \hat{E}^{a_1 i_1}_{a_2 a_3} \tag{175}$$
$$+ f^{i_2}_{i_3} t^{a_1}_{i_2} t^{a_2 a_3}_{i_3 i_1} \hat{E}^{a_1 i_1}_{a_2 a_3} \tag{176}$$
$$- \tfrac{1}{2} D^{i_3}_{i_4} f^{a_3}_{a_4} t^{a_3 a_1}_{i_2 i_3} t^{a_4 a_2}_{i_1 i_4} \hat{E}^{a_1 i_1}_{a_2 i_2} \tag{177}$$
$$+ \tfrac{1}{4} D^{i_3}_{p_1} f^{i_4}_{p_1} t^{a_3 a_1}_{i_2 i_3} t^{a_3 a_2}_{i_1 i_4} \hat{E}^{a_1 i_1}_{a_2 i_2} \tag{178}$$
$$+ \tfrac{1}{4} D^{i_3}_{p_1} f^{i_4}_{p_1} t^{a_3 a_1}_{i_2 i_4} t^{a_3 a_2}_{i_1 i_3} \hat{E}^{a_1 i_1}_{a_2 i_2} \tag{179}$$
$$+ \tfrac{1}{4} D^{i_2}_{i_3} f^{i_4}_{p_1} t^{a_3 a_1}_{i_4 i_2} t^{a_3 a_2}_{i_1 i_3} \hat{E}^{a_1 i_1}_{a_2 p_1} \tag{180}$$
$$- \tfrac{1}{2} f^{a_3}_{p_1} t^{a_1}_{i_2} t^{a_3 a_2}_{i_1 i_2} \hat{E}^{a_1 i_1}_{a_2 p_1} \tag{181}$$
$$- \tfrac{1}{2} D^{i_3}_{i_4} f^{a_3}_{a_4} t^{a_3 a_1}_{i_2 i_3} t^{a_4 a_2}_{i_4 i_1} \hat{E}^{a_1 i_1}_{i_2 a_2} \tag{182}$$
$$- \tfrac{1}{2} D^{i_3}_{i_4} f^{a_3}_{a_4} t^{a_3 a_1}_{i_3 i_2} t^{a_4 a_2}_{i_1 i_4} \hat{E}^{a_1 i_1}_{i_2 a_2} \tag{183}$$
$$+ D^{i_3}_{i_4} f^{a_3}_{a_4} t^{a_3 a_1}_{i_3 i_2} t^{a_4 a_2}_{i_4 i_1} \hat{E}^{a_1 i_1}_{i_2 a_2} \tag{184}$$
$$+ \tfrac{1}{4} D^{i_3}_{p_1} f^{i_4}_{p_1} t^{a_3 a_1}_{i_2 i_3} t^{a_3 a_2}_{i_4 i_1} \hat{E}^{a_1 i_1}_{i_2 a_2} \tag{185}$$
$$+ \tfrac{1}{4} D^{i_3}_{p_1} f^{i_4}_{p_1} t^{a_3 a_1}_{i_2 i_4} t^{a_3 a_2}_{i_3 i_1} \hat{E}^{a_1 i_1}_{i_2 a_2} \tag{186}$$
$$+ \tfrac{1}{4} D^{i_3}_{p_1} f^{i_4}_{p_1} t^{a_3 a_1}_{i_3 i_2} t^{a_3 a_2}_{i_1 i_4} \hat{E}^{a_1 i_1}_{i_2 a_2} \tag{187}$$
$$- \tfrac{1}{2} D^{i_3}_{p_1} f^{i_4}_{p_1} t^{a_3 a_1}_{i_3 i_2} t^{a_3 a_2}_{i_4 i_1} \hat{E}^{a_1 i_1}_{i_2 a_2} \tag{188}$$
$$+ \tfrac{1}{4} D^{i_3}_{p_1} f^{i_4}_{p_1} t^{a_3 a_1}_{i_4 i_2} t^{a_3 a_2}_{i_1 i_3} \hat{E}^{a_1 i_1}_{i_2 a_2} \tag{189}$$
$$- \tfrac{1}{2} D^{i_3}_{p_1} f^{i_4}_{p_1} t^{a_3 a_1}_{i_4 i_2} t^{a_3 a_2}_{i_3 i_1} \hat{E}^{a_1 i_1}_{i_2 a_2} \tag{190}$$
$$- \tfrac{1}{4} D^{i_4}_{p_1} f^{a_2}_{p_1} t^{a_2 a_3}_{i_1 i_4} t^{a_3 a_1}_{i_3 i_2} \hat{E}^{a_1 i_1}_{i_2 i_3} \tag{191}$$
$$+ \tfrac{1}{2} D^{i_4}_{p_1} f^{a_2}_{p_1} t^{a_2 a_3}_{i_4 i_1} t^{a_3 a_1}_{i_3 i_2} \hat{E}^{a_1 i_1}_{i_2 i_3} \tag{192}$$
$$+ f^{a_2}_{a_3} t^{a_2 a_1}_{i_3 i_2} t^{a_3}_{i_1} \hat{E}^{a_1 i_1}_{i_2 i_3} \tag{193}$$
$$- \tfrac{1}{4} D^{i_3}_{i_4} f^{a_2}_{p_1} t^{a_2 a_3}_{i_1 i_3} t^{a_3 a_1}_{i_2 i_4} \hat{E}^{a_1 i_1}_{i_2 p_1} \tag{194}$$
$$+ \tfrac{1}{2} D^{i_3}_{i_4} f^{a_2}_{p_1} t^{a_2 a_3}_{i_1 i_3} t^{a_3 a_1}_{i_4 i_2} \hat{E}^{a_1 i_1}_{i_2 p_1} \tag{195}$$
$$- \tfrac{1}{4} D^{i_3}_{i_4} f^{a_2}_{p_1} t^{a_2 a_3}_{i_3 i_1} t^{a_3 a_1}_{i_4 i_2} \hat{E}^{a_1 i_1}_{i_2 p_1} \tag{196}$$
$$- \tfrac{1}{2} f^{i_3}_{p_1} t^{a_2 a_1}_{i_3 i_2} t^{a_2}_{i_1} \hat{E}^{a_1 i_1}_{i_2 p_1} \tag{197}$$
$$+ \tfrac{1}{4} D^{i_2}_{i_3} f^{i_4}_{p_1} t^{a_3 a_1}_{i_2 i_4} t^{a_3 a_2}_{i_1 i_3} \hat{E}^{a_1 i_1}_{p_1 a_2} \tag{198}$$
$$- \tfrac{1}{2} D^{i_2}_{i_3} f^{i_4}_{p_1} t^{a_3 a_1}_{i_2 i_4} t^{a_3 a_2}_{i_3 i_1} \hat{E}^{a_1 i_1}_{p_1 a_2} \tag{199}$$
$$+ \tfrac{1}{4} D^{i_2}_{i_3} f^{i_4}_{p_1} t^{a_3 a_1}_{i_4 i_2} t^{a_3 a_2}_{i_3 i_1} \hat{E}^{a_1 i_1}_{p_1 a_2} \tag{200}$$
$$- \tfrac{1}{2} f^{a_3}_{p_1} t^{a_1}_{i_2} t^{a_3 a_2}_{i_2 i_1} \hat{E}^{a_1 i_1}_{p_1 a_2} \tag{201}$$
$$- \tfrac{1}{4} D^{i_3}_{i_4} f^{a_2}_{p_1} t^{a_2 a_3}_{i_3 i_1} t^{a_3 a_1}_{i_2 i_4} \hat{E}^{a_1 i_1}_{p_1 i_2} \tag{202}$$

$$- \frac{1}{2} f^{i_3}_{p_1} t^{a_2 a_1}_{i_2 i_3} t^{a_2}_{i_1} \hat{E}^{a_1 i_1}_{p_1 i_2} \tag{203}$$
$$+ \frac{1}{4} D^{i_1}_{i_2} f^{a_4}_{p_1} t^{a_2 a_3}_{i_1 i_3} t^{a_4 a_1}_{i_3 i_2} \hat{E}^{a_1 p_1}_{a_2 a_3} \tag{204}$$
$$+ \frac{1}{4} D^{i_1}_{i_2} f^{a_4}_{p_1} t^{a_2 a_3}_{i_3 i_1} t^{a_4 a_1}_{i_2 i_3} \hat{E}^{a_1 p_1}_{a_2 a_3} \tag{205}$$
$$- \frac{1}{2} f^{a_4}_{p_1} t^{a_2 a_3}_{i_1 i_2} t^{a_4 a_1}_{i_2 i_1} \hat{E}^{a_1 p_1}_{a_2 a_3} \tag{206}$$
$$+ \frac{1}{2} f^{i_1}_{p_1} t^{a_1}_{i_2} t^{a_2 a_3}_{i_2 i_1} \hat{E}^{a_1 p_1}_{a_2 a_3} \tag{207}$$
$$+ \frac{1}{4} D^{i_2}_{i_3} f^{i_4}_{p_1} t^{a_3 a_1}_{i_1 i_2} t^{a_3 a_2}_{i_4 i_3} \hat{E}^{a_1 p_1}_{a_2 i_1} \tag{208}$$
$$- \frac{1}{2} f^{a_3}_{p_1} t^{a_2}_{i_2} t^{a_3 a_1}_{i_1 i_2} \hat{E}^{a_1 p_1}_{a_2 i_1} \tag{209}$$
$$+ \frac{1}{4} D^{i_2}_{i_3} f^{i_4}_{p_1} t^{a_3 a_1}_{i_1 i_2} t^{a_3 a_2}_{i_3 i_4} \hat{E}^{a_1 p_1}_{i_1 a_2} \tag{210}$$
$$- \frac{1}{2} D^{i_2}_{i_3} f^{i_4}_{p_1} t^{a_3 a_1}_{i_2 i_1} t^{a_3 a_2}_{i_3 i_4} \hat{E}^{a_1 p_1}_{i_1 a_2} \tag{211}$$
$$+ \frac{1}{4} D^{i_2}_{i_3} f^{i_4}_{p_1} t^{a_3 a_1}_{i_2 i_1} t^{a_3 a_2}_{i_4 i_3} \hat{E}^{a_1 p_1}_{i_1 a_2} \tag{212}$$
$$- \frac{1}{2} f^{a_3}_{p_1} t^{a_2}_{i_2} t^{a_3 a_1}_{i_2 i_1} \hat{E}^{a_1 p_1}_{i_1 a_2} \tag{213}$$
$$+ D^{p_2}_{p_3} g^{a_2 p_2}_{p_1 p_3} t^{a_2 a_1}_{i_2 i_1} \hat{E}^{a_1 p_1}_{i_1 i_2} \tag{214}$$
$$- \frac{1}{2} D^{p_2}_{p_3} g^{a_2 p_2}_{p_3 p_1} t^{a_2 a_1}_{i_2 i_1} \hat{E}^{a_1 p_1}_{i_1 i_2} \tag{215}$$
$$- \frac{1}{2} f^{i_3}_{p_1} t^{a_2 a_1}_{i_2 i_1} t^{a_2}_{i_3} \hat{E}^{a_1 p_1}_{i_1 i_2} \tag{216}$$
$$+ h^{a_2}_{p_1} t^{a_2 a_1}_{i_2 i_1} \hat{E}^{a_1 p_1}_{i_1 i_2} \tag{217}$$
$$- \frac{1}{2} D^{i_2}_{p_3} g^{a_2 p_2}_{p_1 p_3} t^{a_2 a_1}_{i_2 i_1} \hat{E}^{a_1 p_1}_{i_1 p_2} \tag{218}$$
$$- \frac{1}{2} D^{i_2}_{p_3} g^{a_2 p_2}_{p_3 p_1} t^{a_2 a_1}_{i_1 i_2} \hat{E}^{a_1 p_1}_{i_1 p_2} \tag{219}$$
$$+ D^{i_2}_{p_3} g^{a_2 p_2}_{p_3 p_1} t^{a_2 a_1}_{i_2 i_1} \hat{E}^{a_1 p_1}_{i_1 p_2} \tag{220}$$
$$- \frac{1}{2} D^{i_2}_{p_3} g^{a_2 p_2}_{p_1 p_3} t^{a_2 a_1}_{i_1 i_2} \hat{E}^{a_1 p_1}_{p_2 i_1} \tag{221}$$
$$- g^{i_1 p_1}_{p_2 p_3} t^{a_1}_{i_1} \hat{E}^{a_1 p_1}_{p_2 p_3} \tag{222}$$
$$- f^{a_3}_{i_3} t^{a_1 a_2}_{i_1 i_3} t^{a_3}_{i_2} \hat{E}^{i_1 i_2}_{a_1 a_2} \tag{223}$$
$$- f^{a_3}_{i_3} t^{a_1}_{i_3} t^{a_3 a_2}_{i_1 i_2} \hat{E}^{i_1 i_2}_{a_1 a_2} \tag{224}$$
$$- \frac{1}{4} D^{i_4}_{p_1} f^{a_2}_{p_1} t^{a_2 a_3}_{i_3 i_4} t^{a_3 a_1}_{i_2 i_1} \hat{E}^{i_1 i_2}_{a_1 i_3} \tag{225}$$
$$+ \frac{1}{2} D^{i_4}_{p_1} f^{a_2}_{p_1} t^{a_2 a_3}_{i_4 i_3} t^{a_3 a_1}_{i_2 i_1} \hat{E}^{i_1 i_2}_{a_1 i_3} \tag{226}$$
$$+ f^{a_2}_{a_3} t^{a_2 a_1}_{i_2 i_1} t^{a_3}_{i_3} \hat{E}^{i_1 i_2}_{a_1 i_3} \tag{227}$$
$$+ D^{p_2}_{p_3} g^{a_2 p_2}_{p_1 p_3} t^{a_2 a_1}_{i_2 i_1} \hat{E}^{i_1 i_2}_{a_1 p_1} \tag{228}$$
$$- \frac{1}{2} D^{p_2}_{p_3} g^{a_2 p_2}_{p_3 p_1} t^{a_2 a_1}_{i_2 i_1} \hat{E}^{i_1 i_2}_{a_1 p_1} \tag{229}$$
$$- \frac{1}{2} f^{i_3}_{p_1} t^{a_2 a_1}_{i_2 i_1} t^{a_2}_{i_3} \hat{E}^{i_1 i_2}_{a_1 p_1} \tag{230}$$
$$+ h^{a_2}_{p_1} t^{a_2 a_1}_{i_2 i_1} \hat{E}^{i_1 i_2}_{a_1 p_1} \tag{231}$$
$$+ f^{a_1}_{a_2} t^{a_1 a_3}_{i_1 i_2} t^{a_2 a_3}_{i_3 i_4} \hat{E}^{i_1 i_2}_{i_3 i_4} \tag{232}$$
$$+ \frac{1}{2} f^{a_1}_{p_1} t^{a_1 a_2}_{i_2 i_1} t^{a_2}_{i_3} \hat{E}^{i_1 i_2}_{i_3 p_1} \tag{233}$$
$$- \frac{1}{2} f^{i_4}_{p_1} t^{a_1 a_2}_{i_1 i_2} t^{a_1 a_2}_{i_3 i_4} \hat{E}^{i_1 i_2}_{i_3 p_1} \tag{234}$$
$$+ \frac{1}{2} g^{a_1 a_2}_{p_1 p_2} t^{a_1 a_2}_{i_1 i_2} \hat{E}^{i_1 i_2}_{p_1 p_2} \tag{235}$$
$$- D^{p_2}_{p_3} g^{i_2 p_2}_{p_1 p_3} t^{a_1 a_2}_{i_1 i_2} \hat{E}^{i_1 p_1}_{a_1 a_2} \tag{236}$$
$$+ \frac{1}{2} D^{p_2}_{p_3} g^{i_2 p_2}_{p_3 p_1} t^{a_1 a_2}_{i_1 i_2} \hat{E}^{i_1 p_1}_{a_1 a_2} \tag{237}$$

$$- \tfrac{1}{2}\, f^{a_3}_{p_1} t^{a_1 a_2}_{i_1 i_2} t^{a_3}_{i_2} \hat{E}^{i_1 p_1}_{a_1 a_2} \tag{238}$$
$$- h^{i_2}_{p_1} t^{a_1 a_2}_{i_1 i_2} \hat{E}^{i_1 p_1}_{a_1 a_2} \tag{239}$$
$$- \tfrac{1}{4}\, D^{i_3}_{i_4} f^{a_2}_{p_1} t^{a_2 a_3}_{i_2 i_3} t^{a_3 a_1}_{i_1 i_4} \hat{E}^{i_1 p_1}_{a_1 i_2} \tag{240}$$
$$+ \tfrac{1}{2}\, D^{i_3}_{i_4} f^{a_2}_{p_1} t^{a_2 a_3}_{i_2 i_3} t^{a_3 a_1}_{i_4 i_1} \hat{E}^{i_1 p_1}_{a_1 i_2} \tag{241}$$
$$- \tfrac{1}{4}\, D^{i_3}_{i_4} f^{a_2}_{p_1} t^{a_2 a_3}_{i_3 i_2} t^{a_3 a_1}_{i_4 i_1} \hat{E}^{i_1 p_1}_{a_1 i_2} \tag{242}$$
$$- \tfrac{1}{2}\, f^{i_3}_{p_1} t^{a_2 a_1}_{i_3 i_1} t^{a_2}_{i_2} \hat{E}^{i_1 p_1}_{a_1 i_2} \tag{243}$$
$$- \tfrac{1}{2}\, D^{i_2}_{p_3} g^{a_2 p_1}_{p_2 p_3} t^{a_2 a_1}_{i_2 i_1} \hat{E}^{i_1 p_1}_{a_1 p_2} \tag{244}$$
$$- \tfrac{1}{2}\, D^{i_2}_{p_3} g^{a_2 p_1}_{p_3 p_2} t^{a_2 a_1}_{i_1 i_2} \hat{E}^{i_1 p_1}_{a_1 p_2} \tag{245}$$
$$+ D^{i_2}_{p_3} g^{a_2 p_1}_{p_3 p_2} t^{a_2 a_1}_{i_2 i_1} \hat{E}^{i_1 p_1}_{a_1 p_2} \tag{246}$$
$$- \tfrac{1}{4}\, D^{i_3}_{i_4} f^{a_2}_{p_1} t^{a_2 a_3}_{i_3 i_2} t^{a_3 a_1}_{i_1 i_4} \hat{E}^{i_1 p_1}_{i_2 a_1} \tag{247}$$
$$- \tfrac{1}{2}\, f^{i_3}_{p_1} t^{a_2 a_1}_{i_1 i_3} t^{a_2}_{i_2} \hat{E}^{i_1 p_1}_{i_2 a_1} \tag{248}$$
$$+ \tfrac{1}{2}\, f^{a_1}_{p_1} t^{a_1 a_2}_{i_3 i_2} t^{a_2}_{i_1} \hat{E}^{i_1 p_1}_{i_2 i_3} \tag{249}$$
$$- \tfrac{1}{2}\, f^{i_4}_{p_1} t^{a_1 a_2}_{i_1 i_4} t^{a_1 a_2}_{i_2 i_3} \hat{E}^{i_1 p_1}_{i_2 i_3} \tag{250}$$
$$- \tfrac{1}{2}\, D^{i_2}_{p_3} g^{a_2 p_1}_{p_2 p_3} t^{a_2 a_1}_{i_1 i_2} \hat{E}^{i_1 p_1}_{p_2 a_1} \tag{251}$$
$$+ g^{a_1 p_1}_{p_2 p_3} t^{a_1}_{i_1} \hat{E}^{i_1 p_1}_{p_2 p_3} \tag{252}$$
$$+ \tfrac{1}{2}\, D^{i_1}_{p_3} g^{i_2 p_3}_{p_1 p_2} t^{a_1 a_2}_{i_2 i_1} \hat{E}^{p_1 p_2}_{a_1 a_2} \tag{253}$$
$$- \tfrac{1}{2}\, g^{i_1 i_2}_{p_1 p_2} t^{a_1 a_2}_{i_1 i_2} \hat{E}^{p_1 p_2}_{a_1 a_2} \tag{254}$$
$$- g^{i_1 p_3}_{p_1 p_2} t^{a_1}_{i_1} \hat{E}^{p_1 p_2}_{a_1 p_3} \tag{255}$$
$$+ \tfrac{1}{2}\, g^{a_1 a_2}_{p_1 p_2} t^{a_1 a_2}_{i_1 i_2} \hat{E}^{p_1 p_2}_{i_1 i_2} \tag{256}$$
$$+ g^{a_1 p_3}_{p_1 p_2} t^{a_1}_{i_1} \hat{E}^{p_1 p_2}_{i_1 p_3} \tag{257}$$